

Education

University of Waterloo - BAsC, Systems Design Engineering

Sep 2023 – Present

Experience

Hardware Design Engineering Intern – Arista Networks – Santa Clara, CA

Jan 2026 – Apr 2026

- Optimized SerDes routing strategy for a 52-layer 800GbE OSFP switch card through die-to-die channel loss analysis.
- Captured schematics for multiphase VRMs to meet pluggable optics power load and ASIC voltage ripple requirements.
- Added control plane features for power cycling & fault logging with FPGA engineers based on vendor RFP requirements.
- Designed and evaluated stackup for 28-layer mezzanine cards with PCIe/SerDes routing requirements with fab vendors.

Electrical Engineering Intern – Etched – San Jose, CA

May 2025 – Aug 2025

- Designed schematic and layout in Cadence Allegro for a 16-layer HDI test interposer PCB with an STM32 and high-speed signal breakouts (Ethernet, PCIe 5.0) to perform S-parameter tests and emulate basic ASIC functionality.
- Researched and designed a test PCB replicating an ASIC ballmap to evaluate high DC current stress (~1.2 V, 450 A+), assessing reliability risks such as electromigration and thermal runaway from current crowding around BGA balls.
- Helped reduce IR drop by 39% in select ASIC substrate and interposer regions through ballmap pattern optimization.
- Designed 12-layer vendor-specific load slammer PCBs to sink up to 1500A while optimizing stackup and via fanout.

Electrical Engineering Intern – Waterloo Aerial Robotics Group – Waterloo, ON

Sep 2024 – Dec 2024

- Designed mixed-signal PCB integrating a 12-5V buck converter, LDO, ESP32, and impedance-matched RF transceiver in Altium Designer for an ExpressLRS-based RC plane, supporting up to 6 PWM outputs for servo and ESC control.
- Simulated input filtering on a buck converter with and without high-ESR capacitors using LTspice to prevent transient voltage spikes from source impedance of LiPo batteries and resonance from the use of many MLCC capacitors.
- Designed a USB-C source PCB to arbitrate up to 20V 5A for drone debugging, complying with USB-PD & 3.2 spec.

IT Infrastructure & Operations Intern – Grand & Toy – Vaughan, ON

Jan 2024 – Apr 2024

- Managed 250+ computer users through Active Directory, Group Policies, and Microsoft Management Console.

Student Teams

Team Lead – Waterloo Reality Labs – Waterloo, ON

Jan 2024 – Aug 2025

- Created the world's first collegiate engineering design team developing open-source, hackable VR and AR headsets.
- Architected Varifocal, a custom HMD aimed to solve the vergence-accommodation conflict in VR headsets with eye tracking to approximate focal distance and using voice coils to move lenses for real-time focal length adjustment.
- Implemented a camera-based eye tracker with an ESP32, OmniVision camera, IR LEDs, and FOSS tracking software.

Electrical Lead – Waterloo Aerial Robotics Group – Waterloo, ON

Mar 2025 – Present

- Leading the electrical team in the design, build, and testing of sUAS/UAV systems for the AEAC national competition.
- Architected and designed a custom STM32-based flight controller with CAN, SPI, I2C, barometer and redundant IMUs.
- Designed a compact 3-phase motor driver board with an STSPIN32 MCU to control BLDCs on drones with 3S batteries.
- Optimized H-bridge MOSFET selection with calculations for conducting/switching power loss and max temperature.

Skills

- **Hardware:** Power Electronics, Analog/Digital Design, PCB Layout, Schematic Capture, Simulation & Validation
- **Applications & Tools:** Cadence OrCAD, Allegro, Altium Designer, LTspice, SOLIDWORKS, Onshape, Arduino, Git, Linux
- **Languages & Protocols:** C, C++, Python, HTML, CSS, JavaScript, MATLAB, I2C, SPI, UART, USB-PD, USB 3.2